

Kośćciuszkon 2026 – Honeywell Theme #2

1. Title

Detection of GPS Spoofing Attacks Using Machine Learning

2. Short Description

Conduct background research, select and validate an appropriate dataset, design, implement, and evaluate a solution for GPS spoofing detection. The use of machine learning techniques and a focus on GPS spoofing attacks are encouraged; however, classical methods and broader anomaly detection approaches are also acceptable.

3. Detailed Description

An example of actions to be done is presented below:

- Perform background research on the topic
- Identify available datasets, analyze them, and select an appropriate one
- Review relevant evaluation metrics and define those to be used
- Review, implement and test available algorithms or methods
- Provide evidence supporting which solution performs best

4. Criteria

Following criteria will be used:

1. Problem Understanding

- Clear definition of GPS spoofing/anomaly in the chosen context
- Realistic anomaly/spoofing scenarios described
- Relevant input signals/features identified

2. Dataset Selection & Validation (EDA)

- Dataset size (number of samples) reported
- Class distribution (normal vs anomaly) shown
- Dataset includes both normal and anomalous cases
- Coverage discussed (different conditions/scenarios)
- Data quality checked (missing values, inconsistencies)
- Basic statistics provided (e.g., mean, min/max)
- Visualizations provided (e.g. plots, histograms, time-series)
- A justification for why the selected dataset is suitable

3. Feature Engineering / Understanding

- Meaningful features defined (beyond raw data)
- Each feature briefly justified (why useful)
- Feature distributions or behavior analyzed
- Comparison shown between normal vs anomalous data

4. Method / Solution Selection

- Chosen method(s) described and justified
- At least one baseline or alternative approach considered
- Limitations or assumptions acknowledged

5. Implementation & Experimental Setup

- Clear pipeline (data → processing → detection → evaluation)
- Data properly split (training/testing or equivalent)
- Parameters or configuration described
- Implementation is functional (produces results)

6. Evaluation of the Solution

- Evaluation aspects used (e.g., detection rate, false positives)
- Results presented in tables and/or plots
- Confusion matrix (or equivalent) included if applicable
- Evaluation performed on unseen/test data

7. Comparison of Solutions

- Multiple solutions or configurations compared (if applicable)
- Same metrics used consistently across comparisons
- Best solution selected based on evidence

8. Interpretation of Results

- Results are explained (not only presented)
- Failure cases or weaknesses identified
- Trade-offs discussed (e.g., detection vs false alarms)

9. Presentation Quality

- Clear structure (problem → data → method → results → conclusions)
- Figures/tables labeled and readable
- Work is understandable and reproducible